

# Deterministic Real-Time Analytics of Geospatial Data Streams through ScaleGate Objects

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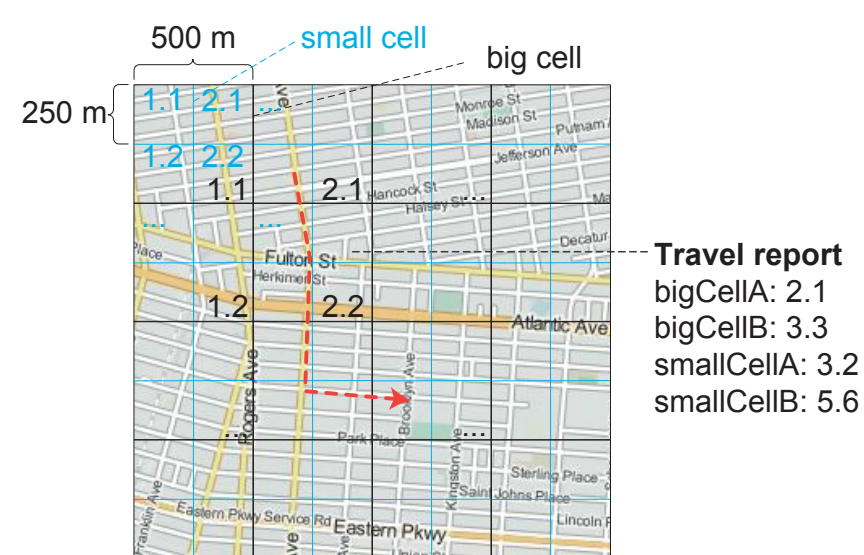
**NOMINATED FOR DEBS 2015 GRAND CHALLENGE AWARD**

## DEBS 2015 GRAND CHALLENGE



Analyze taxi trip reports from NYC and compute:

- Top-10 most frequent routes in the last 30 minutes.
- Top-10 most profitable areas based on the median fare and tip (during the last 15 minutes) and number of empty taxis (during the last 30 minutes)



## OUR APPROACH AND NOVELTY

Scale up, then scale out!

- New pioneering data structures with appropriate API and concurrent implementations, enabling:
- Enhanced Parallel and Distributed Stream Processing Engine's analysis

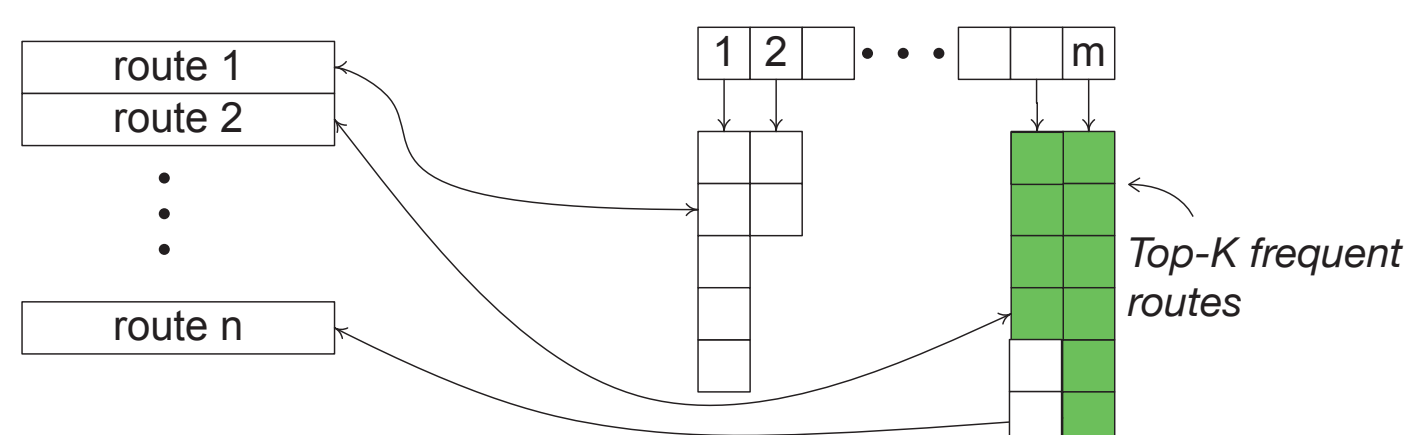
- Enables concurrent and in-order deterministic processing of ready tuples in data streaming
- Lock-free linearizable implementation enables determinism and full fine-grain concurrency

## IMPLEMENTATION

Key data structures maintained by the Processing Units

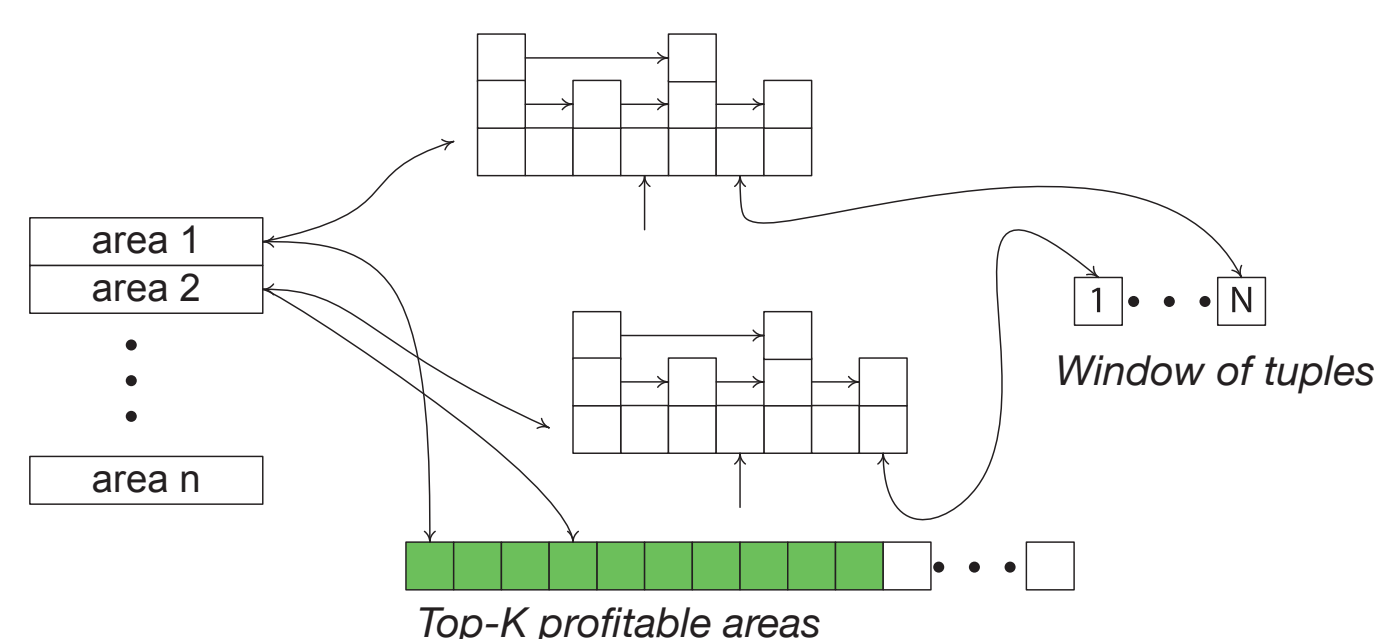
Query 1: Top-K most frequent routes

- Order events' occurrences using routes as unique keys
- Provide Top-K counts in  $O(1)$  time.
- All operations with  $O(1)$  time complexity on average

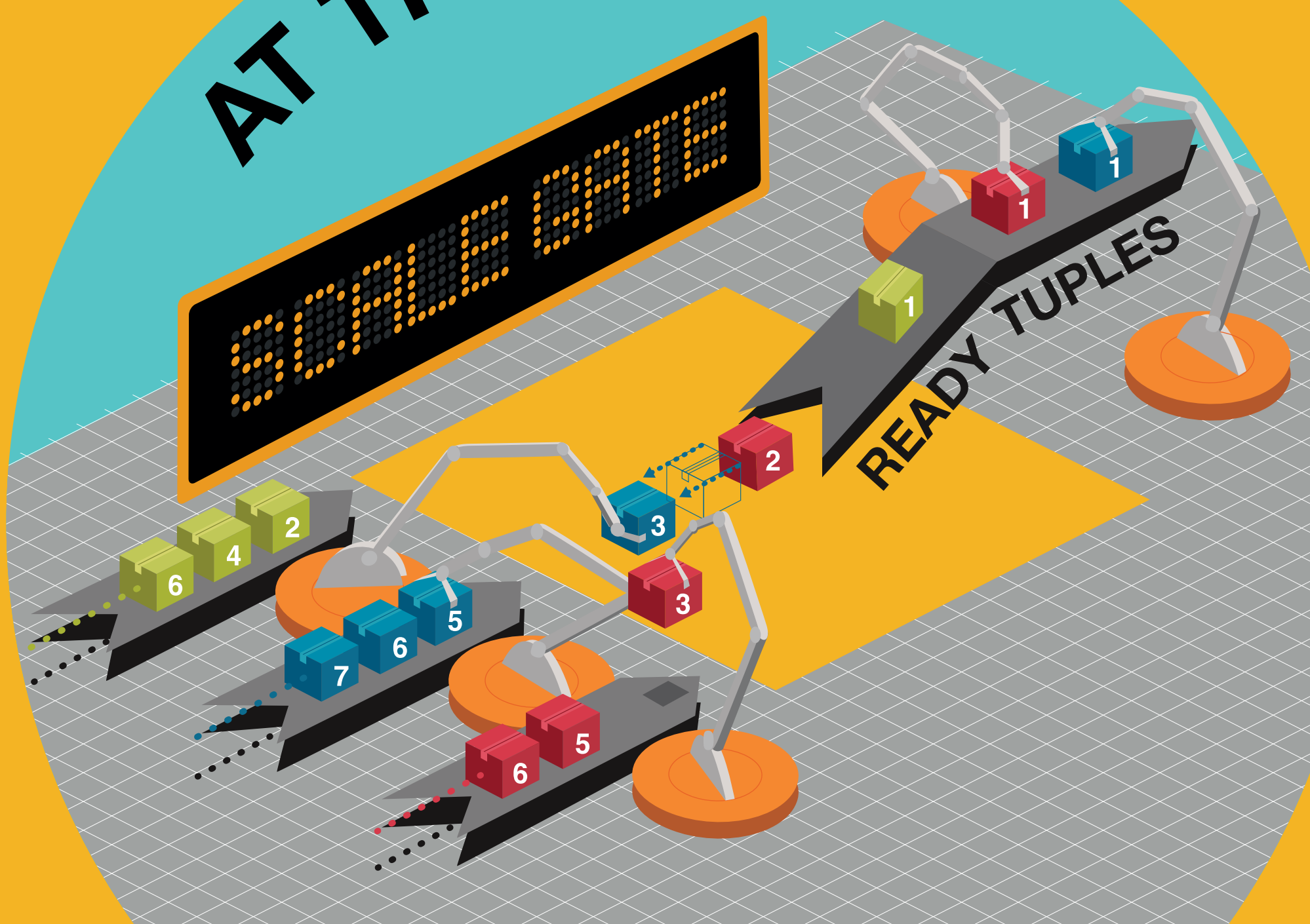


Query 2: Top-K most profitable areas

- Calculate median over a sliding window
  - $O(\log(\text{WinSize}))$  w.h.p. on new tuple
  - $O(1)$  on average on expired tuples
- Maintain PriorityQueue for profitable areas



AT THE CORE:

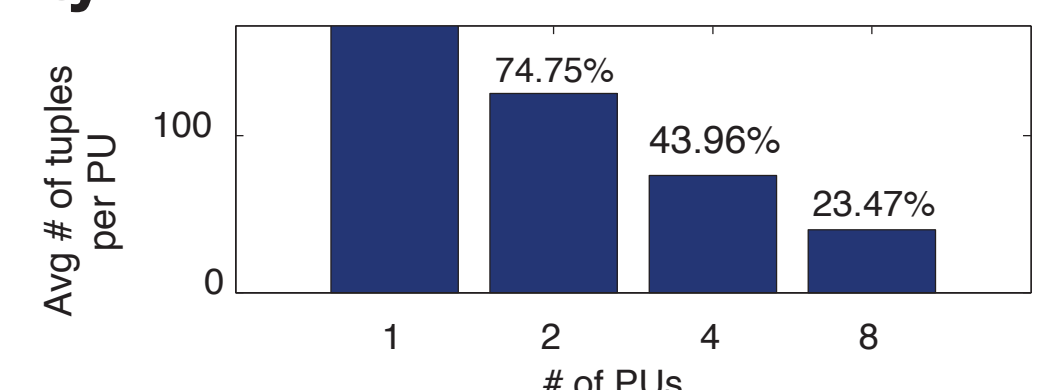


ScaleGate API

`addTuple(timestamp, tuple, sourceID)`  
`getNextReadyTuple(readerID)`

## PERFORMANCE

Applicability



Virtual machine with 4 cores, running on a Intel(R) Xeon(R) CPU E5-2650 0 @ 2.00GHz  
(cache size: 6144 KB)

**Throughput:** 110,000 tuples/second

**Latency:** 46 milliseconds